

assignment12 electromagnetic waves

⚠ This is a preview of the draft version of the quiz

Started: Mar 23 at 6:52pm

Quiz Instructions



Question 1 5 pts

Which EM wave has the highest frequency ?

☐

X-rays

☐

microwaves

☐

γ -rays

☐

IR



Question 2 5 pts

A light source radiates 60.0 W of single-wavelength sinusoidal light uniformly in all directions. What is the amplitude of the magnetic field of this light at a distance of 0.700 m from the bulb? ($\epsilon_0 = 8.85 \times 10^{-12}$

$C^2/N \cdot m^2$, $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$, $c = 3.0 \times 10^8 \text{ m/s}$)

hint: slide 27 give the equation for the EM energy flowing per unit area and unit time S as a function of B_{rms} . Unit is Watt/m² . so $S = \text{total power} / \text{total area}$. Here the total power of the source is 60W source. at a distance 0.7m that energy will spread over the area of a sphere of radius R (0.7m) . So $S = P / (4\pi R^2)$ with $S = (c B^2 / \mu_0)$ solve for B_{rms} . Then multiply by 1.44 to get the amplitude of B . B_{max} .
($B_{max} = B_{rms} \times \sqrt{2}$)

☐

$2.86 \times 10^{-7} \text{ T}$

☐

$2.22 \times 10^{-7} \text{ T}$

☐

$1.76 \times 10^{-7} \text{ T}$

☐

$2.02 \times 10^{-7} \text{ T}$



Question 3 5 pts

An 8.0-mW laser beam emits a cylindrical beam of single-wavelength sinusoidal light 0.90 mm in diameter. What is the rms value of the electric field in this laser beam? ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$, $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$, $c = 3.0 \times 10^8 \text{ m/s}$)

hint: A laser is a peculiar source of light. The light does not spread out spherically so the energy of the laser stays the same regardless of the distance. Like the video I showed in class with I radiation. So the energy per unit area is constant. HERE the beam of light is a cylinder. so the EM waves flow through an area shaped as a circle and not shape as a sphere!. See slide 27. S is the energy per unit area and per unit time (through which the waves go through). is $S = \text{power} / \text{area}$ with area = circle of radius $R = 0.45 \text{ mm}$. Then slide 27 gives you the equation with S as a function of E_{rms} .

☐

2200 N/C

☐

1000 N/C

☐

1100 N/C

☐

2000 N/C

☐

4100 N/C

☐

Question 4 5 pts

An 8.00-mW laser beam emits a cylindrical beam of single-wavelength sinusoidal light 0.600 mm in diameter. What is the maximum value of the magnetic field in the laser beam? ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$, $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$, $c = 3.0 \times 10^8 \text{ m/s}$)

hint: same as in #3. But now you are using the equation between S and B_{rms} . (slide 23). But multiply by 1.44 to get the max.

☐
15.4 μT
☐
17.2 μT
☐
9.24 μT
☐
20.5 μT
☐
12.4 μT
☐

Question 5 5 pts

How much energy is transported across a 1.00-cm^2 area per hour by a sinusoidal electromagnetic wave whose electric field has a maximum strength of 30.4 V/m ? ($\epsilon_0 = 8.85 \times 10^{-12}\text{ C}^2/\text{N} \cdot \text{m}^2$, $\mu_0 = 4\pi \times 10^{-7}$

$\text{T} \cdot \text{m/A}$, $c = 3.0 \times 10^8\text{ m/s}$)

hint: first Find the Erms ($E_{\text{max}}=1.44\text{ Erms}$). Then use slide 27 to find S using Erms.

S is the energy flow per unit area (1m^2) and per unit second.

if you know the energy flowing per second, you find the energy flowing per hour (how many seconds in one hour ?)

if you know the energy flowing per m^2 so you can find for 1 cm^2 (it is less).

$1\text{m} \times 1\text{m} = 100\text{cm} \times 100\text{cm}$

$1\text{ m}^2 = 10,000\text{ cm}^2$ (or $1\text{ cm}^2 = 0.0001\text{ m}^2$ divide by 10,000 both sides)

☐

0.44 mJ

☐

0.44 J

☐

0.44 nJ

☐

0.44 μJ

☐

Question 6 5 pts

We computed in class that the solar radiation we get on Earth is about , per unit area, $1\text{m} \times 1\text{m}$

☐

10W

☐

100W

☐

1000W

☐

1W

☐

Question 7 5 pts

radioisotopes made for PET scan decay in the patient (like the brain) and emit _____ that will be annihilated by electrons. That's how gamma rays are produced inside the person

☐

positrons

☐

protons

☐

electrons



neutrons



neutrinos



Question 8 5 pts

Radio waves that tend to be stopped by large building are



I missed the class



AM



FM



Question 9 5 pts

Electromagnetic waves consist of (lecture)



particles of light energy.



high-frequency sound waves.



compressions and rarefactions of electromagnetic pulses..



oscillating electric and magnetic fields.



Question 10 5 pts

Which one of the following is *not* an electromagnetic wave?



infrared



radio waves



gamma rays



sound waves



ultraviolet



Question 11 5 pts

Which of the following statements about electromagnetic waves in free space are true? (There could be more than one correct choice.)

☐

The electric field vector is always at right angles to the magnetic field vector.

☐

The higher-frequency travel faster than the lower-frequency waves.

☐

The higher-frequency waves have shorter wavelengths than the lower-frequency waves.

☐

The wavelengths of the visible waves are some of the longest electromagnetic waves.

☐

The wavelengths of the visible waves are some of the shortest electromagnetic waves.



Question 12 5 pts

you see the sun

☐

what you talking about

☐

I missed the class

☐

as it was 8 minutes ago

☐

as it is now, at that instant

☐

as it will be 5 minutes from now



Question 13 5 pts

A 7.55×10^{14} Hz electromagnetic wave propagates in carbon tetrachloride with a speed of



$2.05 \times (10)$ with superscript (8) m/s. The wavelength of the wave in carbon tetrachloride is

closest to:

☐

361 nm

☐

272 nm

☐

338 nm

☐

301 nm

☐

397 nm



Question 14 5 pts

How far does a beam of light travel through space in one 365-day year? ($c = 3.0 \times 10^8$ m/s)

☐

36×10^{16} m

☐

30×10^8 m

☐

80×10^{12} m

☐

95×10^{14} m

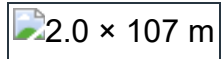
☐

20×10^{15} m

☐

Question 15 5 pts

A radio station broadcasts at 80 MHz. How long does it take for this radio signal to travel a distance of



through space? ($c = 3.0 \times 10^8$ m/s)

☐

20 ms

☐

25 ms

☐

15 ms

☐

6.7×10^{-2} s

☐

0.15×10^{-2} s

☐

Question 16 5 pts

Which one of the following types of electromagnetic wave travels through space the fastest?

☐

infrared

☐

microwaves

☐

radio waves

☐

ultraviolet

☐

They all travel through space at the same speed.

☐

Question 17 5 pts

The energy density of an electromagnetic wave in free space is

- ☐ entirely in the magnetic field.
- ☐ 1/4 in the magnetic field and 3/4 in the electric field.
- ☐ equally divided between the magnetic and the electric fields.
- ☐ 1/4 in the electric field and 3/4 in the magnetic field.
- ☐ entirely in the electric field.



Question 18 5 pts

A certain part of the electromagnetic spectrum ranges from 200 nm to 400 nm. What is the *lowest* frequency associated with this portion of the spectrum? ($c = 3.00 \times 10^8$ m/s)

- ☐ 7.50×10^{13} Hz
- ☐ 7.50×10^{14} Hz
- ☐ 1.50×10^{15} Hz
- ☐ 7.50×10^{15} Hz
- ☐ 1.50×10^{14} Hz



Question 19 5 pts

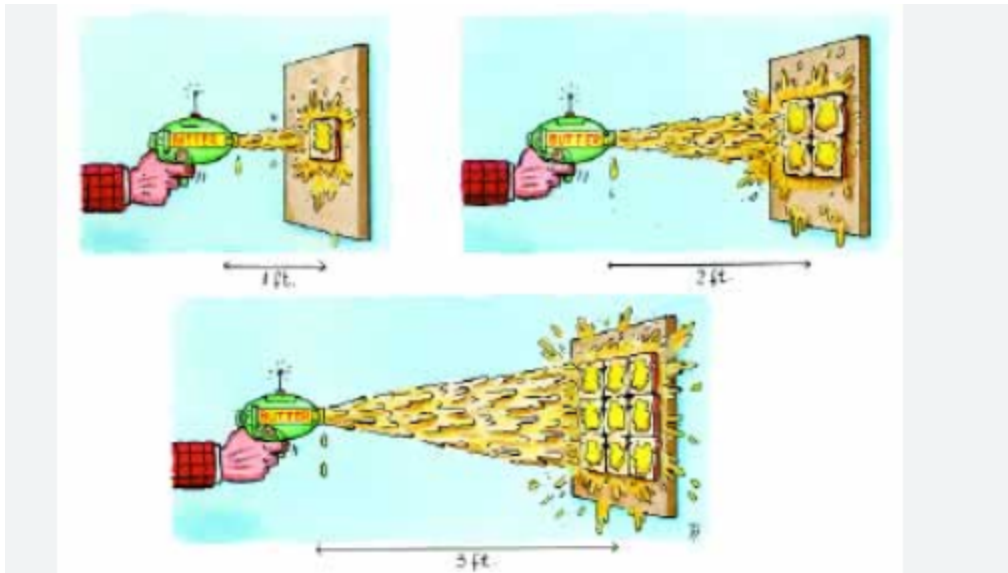
If an electron vibrates up and down 1000 times each second, it generates an electromagnetic wave having a

- ☐ speed of 1000 m/s.
- ☐ period of 1000 s.
- ☐ frequency of 1000 Hz.
- ☐ wavelength of 1000 m.



Question 20 5 pts

A lazy student used a butter gun to spray her French toast at a distance 10cm from the toast placed on a kitchen table. If the distance is 20cm from the table, how many toasts she can spray ? (she has guests over).


☐

4

☐

I missed the class

☐

6

☐

3

☐

2

☒

Question 21 5 pts

Building on the previous question. at a distance of 30cm, she can spray (theoretically)

☐

I missed the class

☐

6 toasts

☐

3 toasts

☐

4 toasts

☐

9 toasts

Not saved

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